

LENISH PANDEY

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Profile

Interdisciplinary researcher, educator, and engineer at the intersection of computational science, mathematics, and STEM outreach. Experienced in machine learning systems, neural simulations, quantum computing, and full-stack development. Equally committed to mentorship, teaching, and creating equitable learning environments. Fluent in bridging technical rigor with clear, human-centered communication.

Education

Florida International University

Aug. 2024 – Apr. 2028 (Expected)

B.S. in Mathematical Data Science – Honors College

Miami, FL

- **Presidential Premier Scholar** | GPA: 3.86/4.0
- **Relevant Coursework:** Data Structures & Algorithms, Machine Learning, Deep Learning, Linear Algebra, Multivariable Calculus, Statistical Physics, Web Development, Calculus I & II, Physics I & II, FIUteach

Research Experience

Pena Lab, Florida Atlantic University

Feb. 2026 – Present

Computational Neuroscience Researcher (via CompNeuroSociety)

Remote – FAU/FIU/FSU Collaboration

- Building and validating a conductance-based **neural circuit simulation** of the *Drosophila* Giant Fiber System in **BRIAN2** and Python, reproducing experimental data on age-associated signal latency.
- Leading technical workshops on **Docker**, **CI/CD pipelines**, **Git/GitHub workflows**, and reproducible research tooling including GitHub Actions and Conda environment management for a multi-institution team (FAU/FIU/FSU).
- Conducting **code reviews** on GitHub PRs; authoring **reproducibility checklists** and technical onboarding documentation; fostering collaborative research culture across institutions.

Department of Physics, Florida International University

Jan. 2025 – May 2025

Research Assistant

Miami, FL

- Collaborated closely with **Dr. Bernard S. Gerstman** to investigate neural network pruning and model efficiency; applied **critical thinking** and **independent research** skills to advance experimental objectives.
- Engineered a modular **mask-pruning pipeline** (0–100%) for **DenseNet-121** in **PyTorch** and **TensorFlow**; developed the **Information Processing Ability (IPA)** metric to benchmark learning efficiency across pruning configurations at scale.
- Designed **reusable, well-documented software components** following **OOP** and clean-code principles; produced **data visualizations and graphics** for research reporting and iterated on findings with the research team.

Incubate Nepal

July 2023 – Sept. 2023

Student Researcher

Kathmandu, Nepal

- Developed a specialized **quantum oracle** for a scheduling optimization problem and programmed quantum circuits using **Qiskit** on IBM Quantum hardware and simulators under the supervision of Dr. Mukesh Ghimire.
- Researched the application of **Variational Quantum Eigensolver (VQE)** to molecular simulation, reviewing 20+ scholarly papers, and authored a research review paper on findings.
- Defended research in front of industry professionals and quantum computing experts; **selected from 600+ applicants** for the competitive research team.

Professional Experience

FuseMachines AI

Dec. 2022 – Jan. 2024

AI Fellowship

Kathmandu, Nepal

- Engineered a custom **OpenAI Gym environment** for the Baghchal board game and trained a **Deep Q-Network** agent using Stable-Baselines3 and TensorFlow, demonstrating end-to-end RL system design.
- Built a complete **carbon emission prediction pipeline** – data ingestion, feature engineering, model training, and evaluation – using Scikit-learn and Pandas.
- Completed intensive AI curriculum spanning **Machine Learning**, **Deep Learning**, **Computer Vision**, and **Natural Language Processing**; earned four micro-degree certifications.

FIUteach Program, FIU

Jan. 2026 – Present

Teaching Intern

Miami, FL

- Competitively selected to design and deliver physical science lessons for middle school students; apply **adaptive teaching, instructional communication**, and **student-centered** strategies across diverse classrooms.
- Collaborate with mentor teachers and FIU faculty to co-develop standards-aligned curricula, applying **critical thinking** and **cross-functional teamwork** to continuously refine lesson delivery.
- Develop **classroom management** skills through structured observation and direct feedback from 2+ supervising educators each semester.

Teaching & Mentorship

Department of Mathematics & Matlab, FIU

Jan. 2025 – Present

Learning Assistant

Miami, FL

- Directly assisted **200+ students** across **3 semesters** and 6 courses (Pre-Calculus through Calculus II and Physics), delivering targeted support through **differentiated instruction** and guided problem-solving.
- Contributed **15 hrs/week** in Matlab, reinforcing comprehension in high-enrollment courses through **active listening**, patience, and clear **one-on-one communication**.
- Partnered with faculty to track student progress, design supplementary problem sets, and uphold academic integrity, demonstrating **professionalism** and **team collaboration**.

Peer Mentor, FIU

Aug. 2025 – Present

Global First Year Experience & Peer Mentor Collective

Miami, FL

- **Global First Year Experience:** Mentored incoming international students on academic planning and campus navigation; facilitated group sessions building community and **peer accountability**.
- **Peer Mentor Collective:** Contributed to a campus-wide mentor network, sharing best practices and participating in professional development training to strengthen outreach and **collaborative support**.
- Maintained consistent communication with program coordinators to align mentoring goals and uphold **professional responsibility** and **organizational standards**.

Student's Research Council Nepal (SRCN)

Sept. 2023 – Dec. 2023

Project Lead

Kathmandu, Nepal

- Spearheaded a 10-week **quantum computing mentorship program** for high school graduates; selected 5 mentees from 150+ applicants and tailored curriculum to diverse academic backgrounds.
- Directed a **multi-disciplinary mentor team**, aligning goals and delivery timelines through clear **delegation, conflict resolution**, and **stakeholder communication**.
- Transformed abstract quantum concepts into accessible lessons; coordinated with QWorld and organized certified workshops with **200+ participants**.

Publications

Solving the General Planar Graph Coloring Problem Using Grover's Algorithm | *IJISRT*

Aug. 2023

- Designed and implemented a full **quantum circuit** in **Qiskit** for the Planar Graph Coloring Problem using Grover's search algorithm; demonstrated a **quadratic speedup** over classical approaches. Published in the International Journal of Innovative Science and Research Technology (IJISRT).

Projects

Data Nexus: Open Data for Social Good | *JavaScript, HTML, CSS, Scikit-learn*

Oct. 2023

- Designed, built, and deployed a **full-stack data aggregation web platform** with 1,500+ categorized datasets across space, climate, and science domains; integrated a **climate prediction ML model** trained on historical CO₂ data with responsive UI and API integration.

DenseNet-121 Pruning Pipeline | *TensorFlow, Keras, PyTorch, Pandas*

Jan. 2025 – Present

- Built a modular **pruning experimentation framework** in Python; abstracted mask generation, evaluation loops, and metric logging into reusable, well-documented components following **OOP** and clean-code principles.

Baghchal Game AI with Deep Q-Network | *Python, TensorFlow, Gymnasium, Stable-Baselines3*

June 2023

- Implemented a **custom Gymnasium environment** from scratch for the two-player Baghchal board game; trained a DQN agent capable of autonomous strategic gameplay.

Leadership & Community

QNepal

July 2023 – Present

General Manager & Mentor

Nepal

- Organized a 1-week certified **quantum computing workshop** with 200+ participants in collaboration with **QWorld**; delivered 2 lectures on quantum computing fundamentals, superdense coding, and quantum teleportation.
- Managed QNepal's LinkedIn and Facebook presence in collaboration with the media team, producing 50+ posts and graphics, achieving a **40% increase in followers**.
- Contributing to a **multi-disciplinary book** on quantum computing applications spanning engineering, CS, physics, business, and policy with MSc & BSc graduates.

Volunteering

South Beach Wine and Food Festival (SOBEWFF)

Feb. 2024

Volunteer – Green Team, Culinary & Event Logistics

Miami, FL

- Contributed **32+ hours** across 3 days, supporting sustainability efforts and coordinating logistics across 3+ venues serving **300+ guests** alongside a team of 20+ volunteers.

Future Educators of America – FIU Chapter

Nov. 2024

Regional Conference Volunteer

Miami, FL

- Partnered with 15+ volunteers to manage registration and guide **200+ high school attendees**, demonstrating **teamwork** and **organizational leadership** throughout the regional conference.

Skills

Languages: Python | C++ | JavaScript | SQL | HTML | CSS | $\text{\LaTeX}$

Frameworks & Libraries: TensorFlow | PyTorch | Keras | Scikit-learn | Gymnasium | Stable-Baselines3 | Hugging Face | OpenCV | NLTK | BRIAN2 | Qiskit

Tools & Platforms: Git | GitHub | Docker | GitHub Actions | Conda | Jupyter | Pandas | Matplotlib | Linux | Figma | Canva | Google Suite

Concepts: OOP | REST APIs | CI/CD | Containerization | Reproducible Research | Reinforcement Learning | Quantum Computing | Neural Simulation

Core Competencies: Leadership | Cross-functional Collaboration | Mentorship | Public Speaking | Critical Thinking | Adaptability | Conflict Resolution | Technical Writing

Certifications

IBM Quantum Challenge 2024 Achievement | Qiskit Global Summer School 2023 – Quantum Excellence | Qiskit Global Summer School 2022 – Quantum Excellence

Machine Learning Micro-degree | Deep Learning Micro-degree | Computer Vision Micro-degree | Natural Language Processing Micro-degree | Quantum Computing for Natural Science | AI for Climate Change